

Dylan Kawalec

Founding Engineer & Solutions Architect | Privacy-Preserving AI Infrastructure & Confidential Compute

Confidential AI Runtimes • TEE-Backed LLM Agents • Zero-Trust Deployment • Blockchain Infrastructure

B.S. Computer Science (WGU) • A.A.S. Computer Programming (MiraCosta)

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• NYC • Open to Remote

Professional Summary

I design and ship production confidential AI systems on Intel TDX/SGX across Azure, AWS, Cloudflare, and Phala Cloud. I specialize in TEE-backed LLM agent orchestration, local private inference pipelines, remote attestation, and secure deployment that moves research into production. At Phala Network, I helped scale confidential-compute adoption across 24+ partner teams, grew open-source visibility from 34→400+ stars, and co-authored the Trustless Cloud Agents specification with the Ethereum Foundation.

Core Competencies

AI & Agent Systems: PyTorch, LLM agent orchestration ([Warp Engine](#)), local private inference pipelines, Evoformer-inspired training, FastAPI serving.

Confidential Compute: Intel TDX/SGX, Phala Cloud/dstack, Fortanix, remote attestation, enclave TLS, zero-trust verification.

Cloud & Deployment: Azure (SGX VMs), AWS, Cloudflare (Workers/CDN), Docker, GitHub Actions, Terraform.

Blockchain & Crypto: Solidity (ERC-8004), zkMIPS, Aleo zkOS, Substrate, MPC, threshold signing, ZK authentication.

Languages & Systems: Python (8+ yr), Rust, C/C++, Go, TypeScript | TLA+/PlusCal, Postgres, Redis.

Selected Builds & Impact

- Shipped [DI](#) (ERC-8004) + [Warp Engine](#) for attested TDX private-agent execution; deployed on Phala Cloud with Cloudflare CDN.
- Built [defe](#) (Rust CLI): Fortanix SGX provisioning, enclave-native TLS, and threshold signing on Azure SGX VMs for Osmosis decentralized front-ends.
- Designed local private AI training and inference tooling (PyTorch, custom Evoformer architecture, C++ verifier prototypes) for zero-trust authentication R&D at [ENI6MA](#) / [Cosmic](#).
- Led [POEMS](#) implementation on AWS (Aleo Best Tech 2023; \$50K grant); designed AIMBC/Yor Bridge MP-HTLC settlement in Solidity.
- Led confidential AI rollout at Phala: deployed TDX private agents via dstack for 24+ partner teams; supported Trustless Cloud Agents adoption (CCC 17/17 vote).

Experience

Head of Developer Relations & Solutions Architect *Phala Network*

May – Dec 2025

- Led confidential AI rollout on Intel TDX via dstack; authored documentation and reference Docker deployments that moved 24+ partner teams to production private-agent systems.
- Grew dstack repo visibility from 34→400+ GitHub stars; co-authored the Trustless Cloud Agents spec (adopted by the Confidential Computing Consortium – 17/17 vote).

Founding Engineer / Principal Architect *Cosmic Inc. | ENI6MA Research*

Summer 2024 – Jan 2026

- Designed zero-trust architecture (TEE-only execution + cryptographic proof boundaries); built Python/C++ verifiers and local private inference pipelines; delivered the provisional patent package that supported a \$168K friends-and-family raise.
- Authored protocol specifications and threat models, then translated them into verification harnesses and deterministic replay tests for partner technical reviews.
- Published public research excerpts and proof previews via [ENI6MA Research](#) and [Cosmic Public Proof](#).

Head of Developer Relations *GOAT Network (BTC L2)*

Jul 2024 – Jan 2025

- Delivered Founders Club MVP and onboarded 100+ founders; authored zkMIPS/EVM docs and ran BTC L2 workshops (BTC Nashville + Staking Summit Singapore).

R&D Engineer *Osmosis Foundation*

May – Jul 2024

- Shipped production Rust CLI ([defe](#)) for Fortanix SGX on Azure; automated enclave TLS, key lifecycle, and threshold signing for decentralized front-ends.

CTO & Principal Engineer *Invo Technologies*

Apr 2023 – Feb 2024

- Built the Invo MVP on AWS; designed a privacy-preserving exchange (Substrate + Aleo zkOS) and led [POEMS](#) (Aleo Best Tech 2023 + \$50K grant).

Earlier: BNB Chain Labs (US Technology Evangelist; first US hire; host, [BNB Podcast](#)) • Findora/Discreet Labs (Head of DevRel; 80+ validators onboarded) • Harmony (Developer Captain/DAO Governor; \$100K funding; 700+ devs) • Tesla (Model 3 Technician) • Zygo (IBF Technician; precision optics, JWST metrology).

Recognition, Credentials & Publications

Awards: ETHGlobal SF 2024 (Ledger 1st, Polygon 1st), Aleo Best Tech 2023, VeChain×EasyA PowerUp (2nd), ETHDenver 2024 Hacker Deathmatch.

Credentials: RareSkills Research Fellow; ZKU Developer; B.S. Computer Science (WGU); A.A.S. Computer Programming (MiraCosta).

[Phala Cloud / dstack Docs](#)
[ENI6MA Research \(public excerpts\)](#)
[DevMos 2024: DFE Workshop](#)

[GOAT Network Protocol Docs](#)
[Cosmic Prototype \(public repo\)](#)
[Host, The Blackbox Podcast](#)